Automated Deep Learning Framework for Microscopic Diagnosis of Malaria Parasitemia in Thin Blood Smears

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Abstract*—*Malaria remains a major global public health concern causing hundreds of thousands of preventable deaths annually. Traditional manual light microscopy of Giemsa-stained thin blood smears is labor-intensive, operator-dependent, and prone to diagnostic error in resource-limited endemic clinics. In this research, we develop an end-to-end deep convolutional neural network framework for automated detection of Plasmodium-infected erythrocytes. We design, train, and evaluate three distinct architectures: an optimized four-stage Custom Deep CNN, a MobileNetV2 inverted-residual architecture, and an EfficientNetB0 compound-scaled model across 27,558 Giemsa-stained cell images from the NIH dataset. The Custom CNN attains 96.84% accuracy, 96.92% sensitivity, and an AUC-ROC of 0.9912. MobileNetV2 provides edge-computing viability with 96.21% accuracy at a 14 ms inference latency per cell. An interactive clinical diagnostic web application is established to support automated point-of-care laboratory screening.

Keywords: Malaria Detection, Convolutional Neural Networks, Deep Learning, MobileNetV2, EfficientNet, Computer Vision, Point-of-Care Diagnostics.

I. Introduction

Malaria is an acute febrile illness caused by protozoan parasites of the genus *Plasmodium*. Early and precise diagnosis is imperative to prevent severe clinical complications. Although rapid diagnostic tests (RDTs) exist, light microscopy of Giemsa-stained thin blood smears remains the clinical gold standard. However, manual microscopy is limited by observer fatigue, inter-observer discrepancy (exceeding 25%), and a critical deficit of skilled hematologists in endemic zones. This paper introduces an automated computer-aided diagnostic framework to standardize and accelerate malaria screening.

II. Materials and Methodology

Dataset: 27,558 single erythrocyte micrographs (13,779 Parasitized, 13,779 Uninfected) from the NIH / Lister Hill National Center for Biomedical Communications.
Preprocessing: Data integrity verification, uniform scaling, intensity normalization to [0.0, 1.0], and stratified splitting (70% train, 15% validation, 15% test).
Data Augmentation: Random rotations (±20°), translations (±10%), shear (0.15), zoom (±15%), and horizontal/vertical flips to eliminate orientation bias.
Model Architectures:
• *Custom CNN:* 4-stage feature extractor with 32, 64, 128, 128 filters, Batch Normalization, Max Pooling, and spatial Dropout (0.25 to 0.50).
• *MobileNetV2:* Lightweight depthwise separable convolutions enabling sub-20ms inference.
• *EfficientNetB0:* Compound scaling network balancing depth, width, and resolution.

Optimization: Binary Cross-Entropy loss minimized with Adam optimizer (initial learning rate 0.001) under EarlyStopping.

III. Experimental Results and Analysis

The models were evaluated on an independent test set of 4,134 unseen images. The Custom CNN achieved the highest overall diagnostic accuracy (96.84%) and AUC-ROC (0.9912). Crucially, sensitivity reached 96.92%, minimizing dangerous false-negative diagnostic errors. MobileNetV2 demonstrated superior computational efficiency, processing single cell images in 14 ms on standard CPU hardware.

Model	Parameters	Accuracy	Sensitivity	Specificity	F1-Score	AUC-ROC	Latency
Custom CNN	1.25 M	96.84%	96.92%	96.76%	96.84%	0.9912	18 ms
MobileNetV2	2.42 M	96.21%	96.05%	96.37%	96.21%	0.9886	14 ms
EfficientNetB0	4.05 M	96.72%	96.80%	96.64%	96.72%	0.9904	29 ms

IV. Clinical Deployment & Conclusion

A full-stack clinical interface was implemented in Python Flask with an interactive glassmorphic web frontend, allowing medical technicians to perform instant cell micrograph analysis with visual triage recommendations. The proposed deep learning framework demonstrates clinical efficacy, speed, and suitability for real-world deployment in malaria-endemic regions.

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